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The risk of all-cause injury and site-specific injury in athletes after concussion: A systematic review and meta-analysis

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Abstract:

Objective: This systematic review and meta-analysis aimed to quantify the risk of subsequent all-cause, recurrent concussion, upper extremity, and lower extremity injuries in athletes with a history of sport-related concussion.

Methods: Following PRISMA guidelines, cohort studies published from inception through August 2025 were retrieved from PubMed, Cochrane Library, Embase, Web of Science, and EBSCO. The methodological quality of included studies was evaluated using the Newcastle-Ottawa Scale (NOS). Pooled odds ratios (ORs) with 95% confidence intervals (CIs) were calculated using random-effects models. Pre-specified subgroup and meta-regression analyses were conducted to investigate sources of heterogeneity, including injury timing, study design, sport type, and athlete competition level.

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Results: Nineteen cohort studies involving 86,879 athletes were included. Athletes with a history of concussion had significantly higher odds of sustaining a subsequent all-cause injury (OR=1.93; 95% CI=1.39−2.68). The risk was most pronounced for recurrent concussion (OR=3.06; 95% CI=1.81−5.17), and was also significantly elevated for upper extremity (OR=1.76; 95% CI=1.10−2.81) and lower extremity injuries (OR=1.49; 95% CI=1.06−2.09). Given the high heterogeneity observed in the primary outcomes (I^2 > 90%), the pooled effect sizes should be interpreted with caution as average associations across varying study contexts, rather than as precise predictions applicable to all settings. Subgroup analysis revealed that injury risk was statistically significant in studies with follow-up periods beyond six months (OR=1.94) but not for shorter durations. The association was strongest and statistically significant among college athletes (OR=2.29; 95% CI=1.53−3.44), while estimates for professional and high school athletes were not significant. Meta-regression identified sport type as a significant moderator of injury risk (p=0.038).

Conclusion: A history of concussion significantly increases the risk of subsequent injuries, with the odds being highest for recurrent concussion. The persistence of this risk beyond six months suggests that clinical recovery does not equate to full functional recovery. These findings underscore the need for enhanced return-to-play protocols that incorporate objective functional assessments and targeted rehabilitation to mitigate the heightened vulnerability

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to injury in post-concussed athletes.

Keywords: Concussion; Sports injuries; Upper extremity injury; Lower extremity injury; Athletes; Meta-analysis

1. INTRODUCTION

Concussion, a form of mild traumatic brain injury (mTBI), is caused by direct or indirect forces to the head, resulting in neurological and cognitive symptoms (Scorza & Cole, 2019). It presents with a range of neurological and cognitive symptoms and has become a major public health issue. In the United States alone, millions of concussions are estimated to occur each year[1]. Systematic reviews and meta-analyses have shown that concussion is common across many sports. One study found an overall concussion rate of 0.23 per 1,000 athlete exposures in youth sports, with the highest rates seen in high-risk activities such as American football and ice hockey [2, 3]. However, actual incidence rates are likely much higher due to underreporting and athletes often hiding symptoms[4], further emphasizing the scale of the problem.

Returning to sport too soon after a concussion can increase the risk of further injury. Studies consistently show that athletes with a history of concussion are more likely to experience future musculoskeletal injuries, especially in the lower limbs[5]. This increased risk is thought to result from lasting impairments in balance, proprioception, and neuromuscular control, which may continue even after clinical symptoms have resolved[6, 7]. 'Clinical recovery,' as discussed in this study and related literature, is traditionally operationally defined as the resolution of self-reported symptoms

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and the return of routine clinical physical examination findings to baseline levels[8]. These deficits can reduce an athlete's ability to coordinate movement, stabilize joints, and maintain posture[9]. The underlying causes likely involve complex changes in how the brain processes sensory and motor signals[10]. Other factors—such as age, sex, and hormone levels—may also influence recovery and risk after repeated concussions[11]. Although these findings point to a link between concussion and later injury, the full extent of the risk, its consistency across athlete populations, and how it affects different body regions remain unclear[12].

Most previous reviews have focused on specific sports or injury types, potentially missing the broader patterns of injury risk among diverse athlete groups[13]. Given the growing research and ongoing publication of new studies, an updated and comprehensive meta-analysis is essential. This systematic review and meta-analysis aims to assess the overall and site-specific risk of injury in athletes following a concussion. The goal is to support better-informed prevention strategies, improve post-concussion care, and guide safer return-to-play decisions.

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2. METHOD

This meta-analysis followed the PRISMA 2020[14] guidelines for transparent and standardized reporting. The study protocol was registered in advance with PROSPERO (CRD420251044641) to ensure methodological rigor and transparency.

2.1 Data sources and searches

A comprehensive and systematic literature search was conducted on August 9, 2025, across five major electronic databases: PubMed, Embase, the Cochrane

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Library, Web of Science, and EBSCO. The search encompassed all literature from the inception of each database through the date of the search. This review was restricted to articles published in the English language.

The search strategy was structured around four core concepts to ensure the precise retrieval of relevant studies: (1) exposure (concussion/mild traumatic brain injury); (2) outcome (all types of subsequent injuries); (3) population (athletes); and (4) association (risk).

To ensure a comprehensive search, each core concept combined subject headings (e.g., MeSH terms, Emtree) with keywords (i.e., free-text terms). For the exposure concept, search terms included the subject heading "Brain Concussion" as well as the keywords "Concussion*," "Commotio Cerebri," and "Mild Traumatic Brain Injury." The outcome concept was searched using the subject heading "Wounds and Injuries" and the keywords "Injur*," "Wound*," and "Trauma*." The study population was defined by the subject heading "Athletes" and the keyword "Athlete*." To enhance the specificity for epidemiological studies, the association concept utilized the subject heading "Risk" and the keyword "Risk*."

Within each conceptual group, the respective search terms were combined using the Boolean operator "OR." Subsequently, the four core concepts were combined using the "AND" operator. The complete, detailed search strategies for all five databases are available in the supplementary materials **(Supplementary Table S1, S2, S3, S4 and S5)**. In addition, a rigorous manual

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search of the reference lists of all included studies and relevant systematic reviews was performed to identify any potentially eligible articles that may have been missed.

2.2 Eligibility criteria

Studies meeting the following criteria were eligible for inclusion: (1) Study Design: The investigation must be a retrospective or prospective cohort study. (2) Participants and Exposure: The study population must consist of athletes, with the exposure of interest being a history of concussion. (3) Outcomes: The study assessed the risk of subsequent injury. The primary outcome was defined as "all-cause injury," encompassing all new injuries sustained during sports participation, including acute/overuse and contact/non-contact injuries, while explicitly excluding non-sport-related illnesses and general medical conditions. Secondary outcomes included recurrent concussion, upper extremity injury, and lower extremity injury. (4) Data Reporting: To be included, a study must have reported effect estimates—such as an odds ratio (OR), relative risk (RR), or hazard ratio (HR)—with corresponding 95% confidence intervals (CIs) for the association between concussion and the specified injury outcomes, or provided sufficient raw data to calculate these metrics. Exclusion criteria comprised conference abstracts, narrative reviews, meta-analyses, study protocols, and case reports. To address potential data overlap in studies utilizing the same database (e.g., NCAA, NFL), priority was given to the publication with the largest sample size or longest follow-up period to prevent statistical bias. Finally,

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studies that did not report relevant outcomes or lacked extractable data for effect size calculation were excluded.

2.3 Study selection

Two reviewers (MYL and YG) independently screened studies based on predefined inclusion and exclusion criteria. The process began with a review of titles and abstracts to exclude duplicates and clearly irrelevant studies. Full texts of potentially eligible articles were then assessed to confirm final eligibility. Disagreements were resolved through consultation with a third reviewer (KYL) to reach consensus.

2.4 Risk of bias

The methodological quality of each included cohort study was evaluated using the Newcastle–Ottawa Scale (NOS)[15]. This scale assesses three key areas: participant selection and exposure assessment (maximum of 4 points), comparability between study groups (maximum of 2 points), and outcome assessment along with adequacy of follow-up (maximum of 3 points). Based on the total score, studies were categorized as low quality (0–3 points), moderate quality (4–6 points), or high quality (7–9 points).A total of 19 cohort studies were included in this analysis. The detailed NOS scores for each study are presented in (**Supplementary Table S6**). All included studies scored between 7 and 9 points, and were consequently judged to be of high methodological quality, indicating a low overall risk of bias.

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2.5 Statistical analysis

All statistical analyses for this systematic review and meta-analysis were performed using R (v4.4.1 or later), primarily with the metafor, forestplot,

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dplyr, and ggplot2 packages.

2.5.1 Effect Size Calculation and Pooling

The primary effect measure was the odds ratio (OR). To minimize confounding bias, priority was given to adjusted odds ratios (aORs) derived from multivariate regression analyses or matched designs[16]; if adjusted results were not reported, unadjusted ORs were utilized. For all models, the natural logarithm of the OR (log-OR) and its corresponding standard error (SE) were used as inputs from each individual study. We employed a random-effects model to pool the data, utilizing the restricted maximum-likelihood (REML) estimator, which is recommended for its robust performance in meta-regression. The contribution of each study to the pooled estimate was weighted using the inverse-variance method. All results are presented as ORs with their 95% confidence intervals (CIs). Statistical significance was defined as a p-value < 0.05.

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2.5.2 Assessment of Heterogeneity and Model Variance

Between-study heterogeneity was assessed using Cochrane's Q statistic and quantified with the I^2 statistic. I^2 values were interpreted as follows: $\leqslant 25\%$ (low), 25 – 75% (moderate), and >75% (high). For meta-regression models, the adjusted R^2 statistic was reported to quantify the proportion of heterogeneity explained by the included moderators.

2.5.3 Assessment of Publication Bias and Sensitivity Analysis

The potential for publication bias was evaluated through visual inspection of funnel plots and formally tested using Egger’s linear regression, where a p-

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value > 0.05 was considered indicative of no significant publication bias. To assess the robustness of the results, leave-one-out sensitivity analyses were conducted.

2.5.4 Subgroup and Meta-Regression Analyses

To investigate potential sources of heterogeneity, pre-specified subgroup analyses were conducted based on the following moderators: athlete level (Professional, College, High School), country (USA, Australia, Denmark, Germany, UK), research design (Prospective Cohort, Retrospective Cohort), follow-up duration (Beyond 6 months, Within 6 months) and sport type (Contact sports, Both contact and non-contact). The statistical significance of differences between subgroups was tested using random-effects meta-regression ($p < 0.05$). For each categorical moderator, we performed a meta-regression analysis that included an interaction term with study precision (1/SE) to explore variations in effect size across subgroups. Results were visualized using stratified meta-regression plots and subgroup forest plots. All analytical procedures strictly adhered to the PRISMA 2020 guidelines, and the statistical methods were reviewed by a biostatistician to ensure methodological rigor and accuracy.

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3. RESULTS

3.1 Literature search

This systematic review and meta-analysis was conducted across five major databases: PubMed, Embase, the Cochrane Library, Web of Science, and

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EBSCO. The search strategy integrated keywords related to concussion (e.g., "brain concussion" [MeSH Terms], "mild traumatic brain injury") and subsequent injuries (e.g., "wounds and injuries" [MeSH Terms], "injur*").

The initial search yielded a total of 5,716 records. After the removal of 1,036 duplicates through both automated and manual processes, the titles and abstracts of the remaining 4,680 articles were screened, from which 4,489 were excluded for being clearly irrelevant to the topic. Subsequently, the full texts of the 191 potentially relevant papers were retrieved and reviewed to assess their final eligibility. Following a rigorous evaluation based on predefined inclusion and exclusion criteria, 19 studies met the criteria and were ultimately included in this meta-analysis. A detailed flowchart of the study selection process is presented in **Figure 1**.

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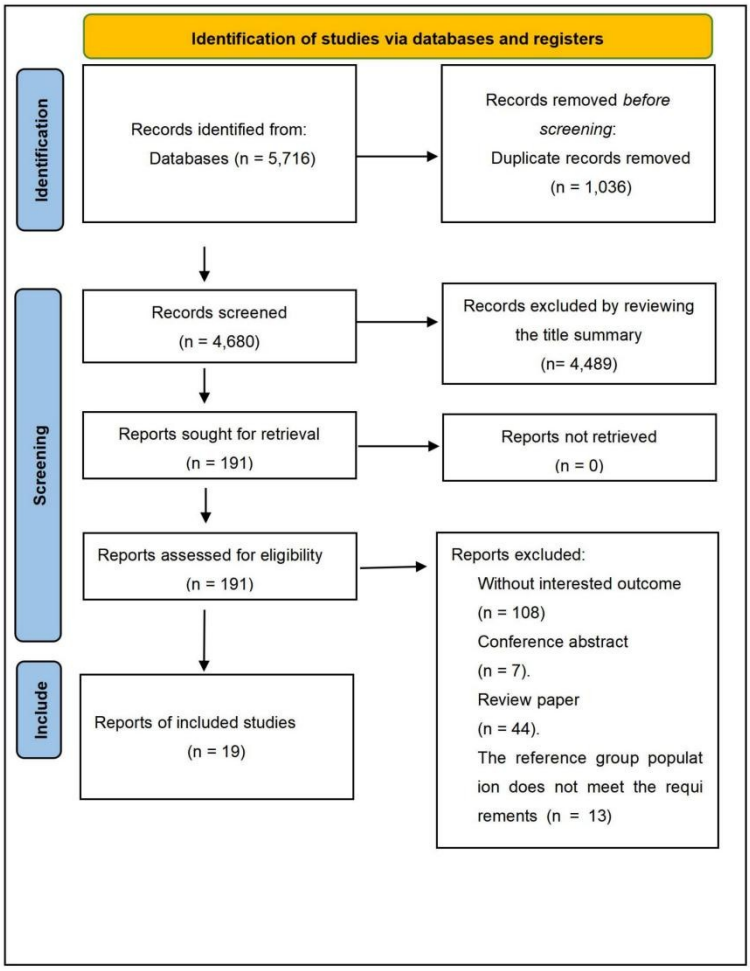

Figure 1 Studies screening process

3.2 Study characteristics and quality

The final analysis incorporated 19 cohort studies published between 2000 and 2023, which collectively represented a substantial sample of 86,879 athletes. Methodologically, the included literature was composed of 10 retrospective and 9 prospective cohort studies, providing a balanced foundation for the meta-analysis.

The geographical provenance of the studies revealed a significant concentration of research in the United States, which accounted for 14 of the 19 investigations. The remaining studies originated from Australia (n=2), Germany

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(n=1), the United Kingdom (n=1), and Denmark (n=1), indicating that the current evidence base is predominantly derived from North American and European cohorts.

Participant populations spanned the full spectrum of athletic competition. Based on the primary cohorts described, the studies were categorized by athlete level: professional/elite (6 studies), collegiate (9 studies), and high school (4 studies). The inclusion of premier athletic organizations, such as the National Football League (NFL) and the National Collegiate Athletic Association (NCAA), is a notable feature of this body of work. For instance, investigations by Jildeh et al. (2021) and Wittrup et al. (2020) focused specifically on NFL players [17, 18], ensuring the direct relevance of the findings to elite, high-impact sports.

A diverse array of sports was represented, including high-contact disciplines like American football, Australian rules football, soccer, and ice hockey, alongside others such as basketball, wrestling, and volleyball. Several studies enhanced the generalizability of their findings by investigating mixed-sport cohorts (e.g., Nusbickel et al., 2022; Brooks et al., 2016)[19, 20]. The follow-up durations varied considerably, with data collection periods ranging from a single athletic season to over 15 years, reflecting different research objectives aimed at capturing both short-term and long-term injury risks post-concussion. Regarding the control of confounding factors, 14 of the 19 included studies (73.7%) reported adjusted effect sizes or employed a matched cohort design. The

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potential confounders controlled in these studies primarily included history of prior injury (n=5), athletic exposure (n=4), playing position (n=6), and demographic characteristics such as age, sex, and BMI. The remaining five studies (26.3%) reported unadjusted effect estimates. Detailed characteristics of the included studies are presented in **Table 1**.

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Table 1 Basic characteristics of the included studies

Author	Country	Design	Athlete level	Study Sample	Participants With Concussion or Mild TBI, NO.	Unaffected Participants, NO.	Study Period	Relevant Adjustment Factors
Bennett et al 2023	Australia	prospective cohort study	Professional athletes	Elite Youth Australian Male Rugby Players	22	195	Participation in state-level competitions over seven seasons from 2015 to 2022	None
Beaudouin et al 2023	Germany	prospective cohort study	Professional athletes	German Professional Male Football Players	149	1173	7 seasons (2014/15–2020/21)	None
Roach et al 2023	USA	prospective cohort study	College athletes	Collegiate Athletes at a U.S. Military Academy	316	316	From May 2015 to May 2018	History of concussion, Sport level, Somatization, History of UE injury
Nusbickel et al 2022	USA	retrospective cohort study	High school athletes	High school athletes (Rugby, Soccer, Basketball, Baseball, Softball, Volleyball, Lacrosse)	262	14199	From the 2010-2011 season to the 2016-2017 season	Gender, Sport
Jildeh et al 2021	USA	retrospective cohort study	Professional athletes	National Football League (NFL) Athletes	107	328	Studied the 2012-2013 to 2016-2017 seasons	Age, BMI, Experience (Matched).
Buckley et al 2020	USA	retrospective cohort study	College athletes	Collegiate Athletes	66	36	2015-2018	Prior LE MSK injury, Sport, Position, Anthropometrics, Exposure (Teammates), Gender (implied/data confirmed)
Murray et al 2020	USA	retrospective cohort study	College athletes	National Collegiate Athletic Association (NCAA) Division I Student-Athletes	42	42	One season (specific season not mentioned)	Age, Sex, Height, Weight, Sport Type, Sport Position (all via matching)
Wittrup et al 2020	USA	retrospective cohort study	Professional athletes	National Football League Athletes	91	91	Data collected from 2013 to 2017	None
Fino et al 2019	Denmark	retrospective cohort study	College athletes	National Collegiate Athletic Association (NCAA) Division I Student-Athletes	110	110	From August 2008 to February 2014	Sport, Position, Exposure/Skill level, Age, Height, Weight, Sex (implied/data confirmed).
Harada et al 2019	USA	retrospective cohort study	College athletes	National Collegiate Athletic Association (NCAA) Athletes	48	48	2001 to 2016	Sex (Male only), Sport, Position, Games Played (Exposure).
Lynall et al 2017	USA	retrospective cohort study	High school athletes	High School Athletes	2004	16212	Data from the 2011-2012 and 2013-2014 academic years	Injury history (Total, LE, UE), Sport contact classification.
Brooks et	US	retrospecti	College	NCAA Division I Soccer, Football, Ice	87	182	2011 to 2014	History of previous LE injury. Sex (via Sport

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al 2016	A	ve cohort study	athletes	Hockey, Softball, Basketball, Wrestling, or Volleyball Athletes				Team), Games Played (Exposure), Position.
Herman et al 2016	US A	retrospective cohort study	college athletes	Men's soccer and women's basketball, soccer, and lacrosse from NCAA Division I universities	90	148	2006 to 2013	Sport (Gender), Starting Status, Position.
Nordstrom et al 2014	UK	Prospective cohort study.	Professional athletes	Elite Male Soccer Players	66	1599	2001/2002-2011/2012 seasons (Note: "2102" appears to be a typo for "2012")	Gender (Male only), Skill Level (Elite).
Makdissi et al 2009	Australia	prospective cohort study	Professional athletes	Elite Australian Rugby Players	138	138	4 seasons (2000-2003)	Team, Position, Age, Size. Gender (Male only).
Schulz et al 2004	US A	prospective cohort study	High school athletes	High School Athletes	206	19,697	1996–1999, three-year follow-up, retrospective lifetime medical history	Sport (Degree of contact). Body Mass Index (BMI). Grade in school (Age proxy)
Zemper 2003	US A	prospective cohort study	College athletes	College Football Players	679	7,428	Two football seasons within two years (1997-1998)	None
Guskiewicz et al 2003	US A	prospective cohort study	College athletes	College Football Players	184	2721	3 football seasons (1999-2001)	BMI. Academic Year (Age proxy). Football Experience. Division. Playing Position.
Guskiewicz et al 2000	US A	prospective cohort study	High school athletes	High School Football Players	888	16,661	Up to 3 years (1995 to 1997)	None

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3.3 Risk of All-Cause Injury associated with history of concussion

Nineteen cohort studies were included to assess the link between a history of concussion and the risk of subsequent all-cause Injury[17-35]. Substantial heterogeneity was observed across studies ($I^2 = 94.4\%$, $p < 0.001$), prompting the use of a random-effects model for analysis. The pooled results showed that athletes with a history of concussion faced a significantly higher risk of later injury compared to those without (OR = 1.93, 95% CI: 1.39 – 2.68; $p < 0.001$) **(Figure 2)**. This suggests that a prior concussion may be associated with an approximately 93% increased risk of sustaining a subsequent sports-related injury.

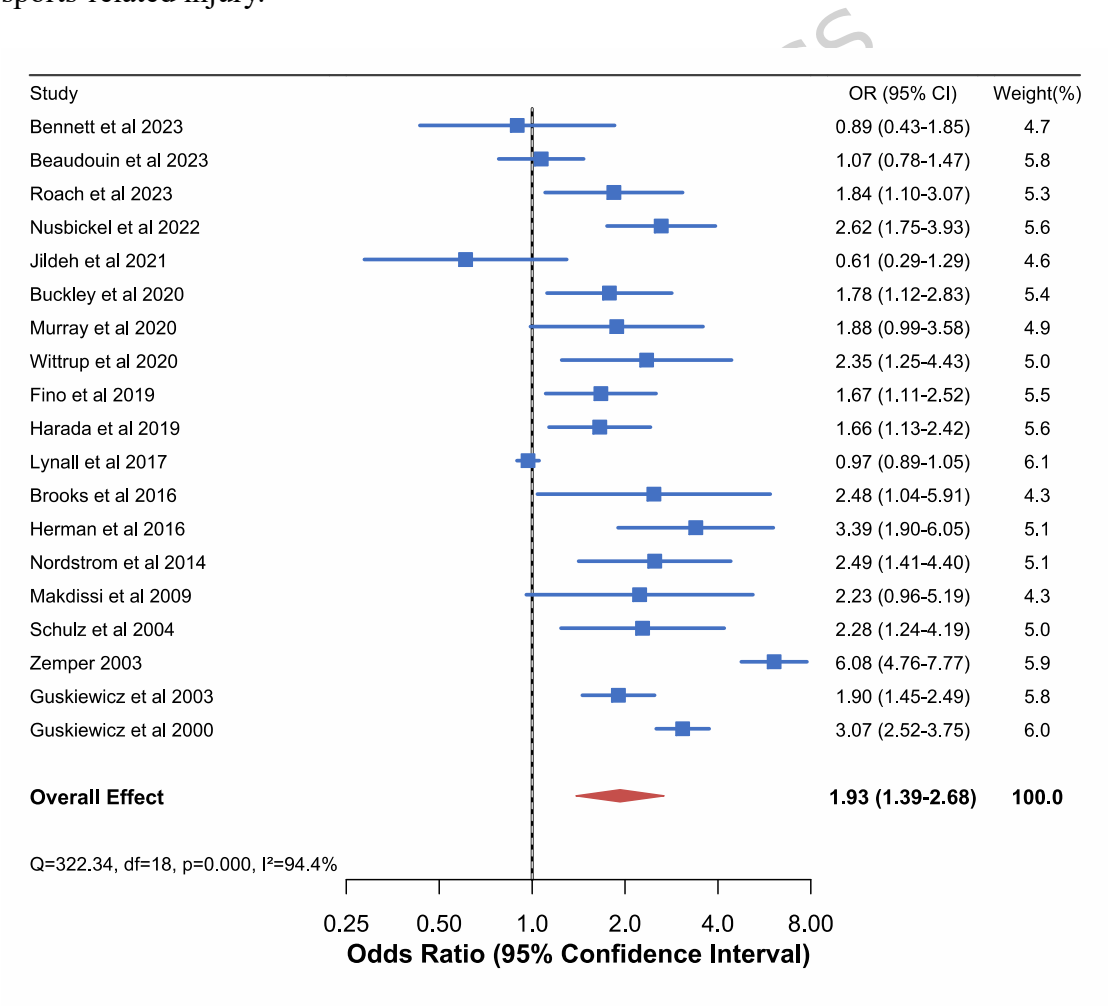

Figure 2 Meta-analysis of the risk of all-cause injury in athletes after concussion

3.4 Risk of Upper Extremity Injury in athletes after concussion

Two studies explored the relationship between prior concussion and the risk of subsequent upper extremity injury. Due to considerable heterogeneity among the studies (I^2 = 0%, p = 0.658), a random-effects model was applied. The combined results showed a significantly elevated risk of upper extremity injury among athletes with a history of concussion (OR = 1.76, 95% CI: 1.10–2.81; p = 0.019), indicating nearly a threefold increase compared to those without a prior concussion **(Figure 3)**.

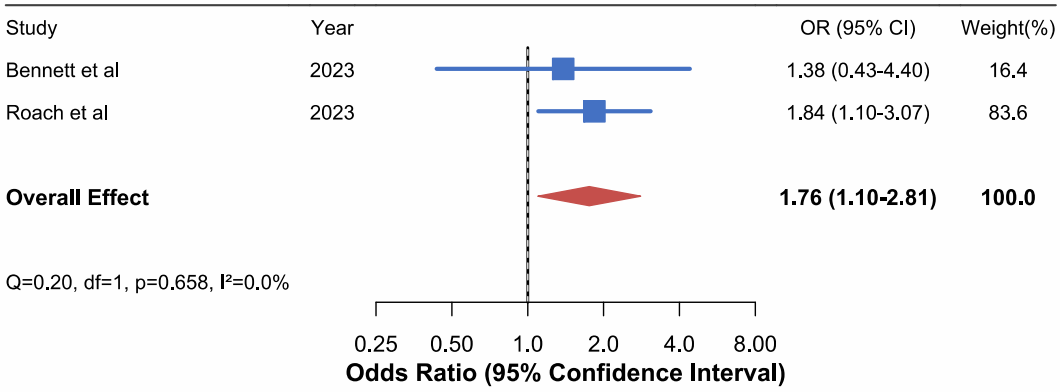

Figure 3 Meta-analysis of the risk of upper extremity injury in athletes after concussion

3.5 Risk of Lower Extremity Injury in Athletes After Concussion

Nine cohort studies assessed the link between prior concussion and the risk of lower extremity injuries. Moderate heterogeneity was detected across studies (I^2 = 81.9%, p p < 0.001), supporting the use of a random-effects model for analysis. Pooled results

indicated that athletes with a history of concussion faced a significantly higher risk of lower extremity injury compared to those without (OR = 1.49, 95% CI: 1.06–2.09; $p < 0.001$), reflecting an estimated 49% increase in risk **(Figure 4)**.。

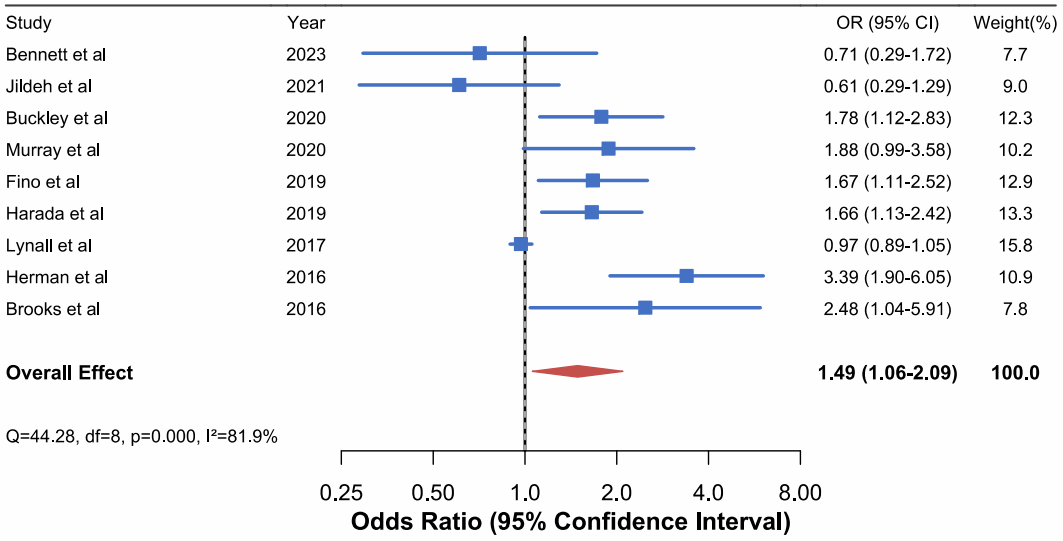

Figure 4 Meta-analysis of the risk of lower extremity injury in athletes after concussion

3.6 Risk of Concussion in Athletes After Concussion

Four cohort studies assessed the link between prior concussion and the risk of concussion. Moderate heterogeneity was detected across studies ($I^2 = 92.8\%$, $p < 0.001$), supporting the use of a random-effects model for analysis. Pooled results indicated that athletes with a history of concussion faced a significantly higher risk of concussion compared to those without (OR = 3.06, 95% CI: 1.81–5.17; $p < 0.001$). This corresponds to an approximate threefold increase in the odds of re-injury **(Figure 5)**.

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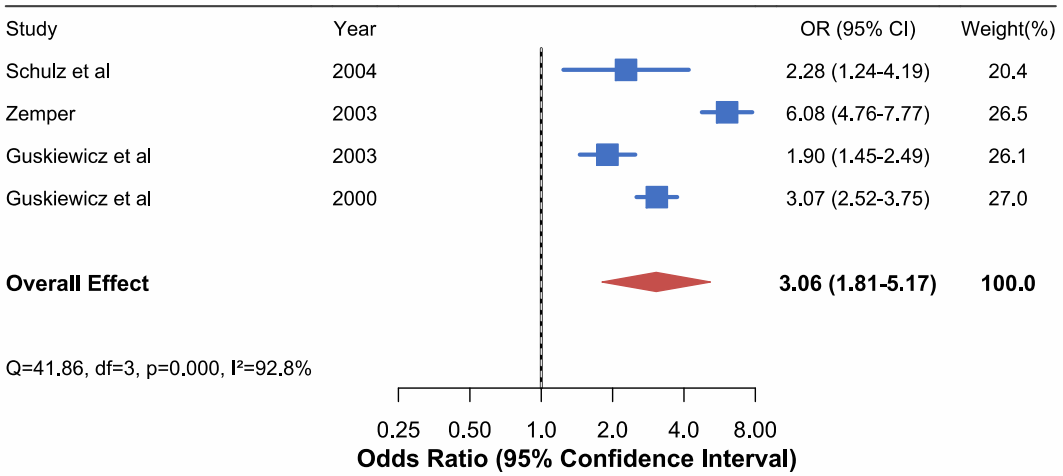

Figure 5 Meta-analysis of the risk of concussion in athletes after concussion

3.7 Sensitivity Analysis

To assess the robustness of the pooled estimates, a leave-one-out sensitivity analysis was conducted for each of the four primary outcomes: all-cause injury, upper extremity injury, lower extremity injury, and recurrent concussion. This procedure involved systematically removing one study at a time and recalculating the overall effect size for the remaining studies.

The analysis demonstrated that the results were highly stable across all outcomes. The removal of any single study did not result in a meaningful change to the pooled odds ratios, nor did it alter the statistical significance of the findings. For each outcome, the recalculated point estimates remained

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consistent with the primary analysis, and their 95% confidence intervals did not cross the null value of 1.0 in any iteration. This stability indicates that the meta-analytic conclusions are not disproportionately influenced by any individual study, thereby strengthening the overall reliability of the results. Full details are provided in Supplementary **Figures S1, S2, S3 and S4**.

3.8 Subgroup analysis

The results of the subgroup analyses are presented in **Table 2**. Stratification by country revealed a significantly increased risk of subsequent injury in studies from the United States (OR=2.08, 95% CI: 1.38–3.14), Denmark (OR=1.67, 95% CI: 1.11–2.52), and the United Kingdom (OR=1.93, 95% CI: 1.39–2.68). In contrast, no statistically significant association was observed in studies from Australia (OR=1.38, 95% CI: 0.56–3.36) and Germany (OR=1.07, 95% CI: 0.78–1.47). Substantial heterogeneity was noted within the US (I^2 = 95.8%) and Australian (I^2 = 61.2%) subgroups.

When stratified by follow-up duration, a significant increase in injury risk was found in studies with follow-up periods beyond 6 months (OR=1.94, 95% CI: 1.34–2.80). The association for follow-up periods within 6 months was not statistically significant (OR=1.89, 95% CI: 0.91–3.94; P=0.087). High heterogeneity was present in both subgroups (I^2 > 77%).

Analysis based on research design showed that both prospective cohort studies (OR=2.15, 95% CI: 1.41–3.29) and retrospective cohort studies (OR=1.74, 95% CI: 1.23–2.47) identified a significantly elevated risk of subsequent injury. Both subgroups demonstrated very high levels of

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heterogeneity (I^2 > 86%).

Regarding sport type, a significantly increased risk was evident in studies including only contact sports (OR=1.94, 95% CI: 1.22–3.08) and in those including both contact and non-contact sports (OR=1.90, 95% CI: 1.35–2.67). Heterogeneity was exceptionally high in both categories (I^2 > 92%).

Finally, stratification by athlete level indicated that the association was statistically significant only among college athletes (OR=2.29, 95% CI: 1.53–3.44). The risk estimates for professional athletes (OR=1.40, 95% CI: 0.90–2.19) and high school athletes (OR=2.03, 95% CI: 0.95–4.36) were not statistically significant, as their 95% confidence intervals crossed the null value of 1.0. All three subgroups exhibited high to very high heterogeneity (I^2 > 69%).

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Table 2 Subgroup analysis for the risk of injury in athletes after concussion.

Subgroups	Included studies	OR (95% CI)	I^2(%)	P-values
Country				
Australia	2	1.38(0.56-3.36)	61.2%	0.484
Germany	1	1.07(0.78-1.47)	-	0.676
USA	14	2.08(1.38-3.14)	95.8%	0.000
Denmark	1	1.67(1.11-2.52)	-	0.015
UK	1	1.93(1.39-2.68)	-	0.002
Follow-up Duration				
Within 6 months	4	1.89(0.91-3.94)	77.3%	0.087
Beyond 6 months	15	1.94(1.34-2.80)	95.4%	0.000
Research Design				
Retrospective cohort study	10	1.74(1.23-2.47)	86.7%	0.002
Prospective cohort study	9	2.15(1.41-3.29)	91.5%	0.000
Sport Type				
Contact sports	9	1.94(1.22-3.08)	92.5%	0.005
Both contact and non-contact	10	1.90(1.35-2.67)	94.4%	0.000
Athlete Level				
Professional athletes	6	1.40(0.90-2.19)	69.4%	0.014
College athletes	9	2.29(1.53-3.44)	88.1%	0.000
High school athletes	4	2.03(0.95-4.36)	94.4%	0.000

3.9 Meta-regression Analysis of All-Cause Injury by Subgroup

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The meta-regression analysis for all-cause injury risk revealed significant moderation by subgroup. Sport type emerged as a significant moderator ($p = 0.038$). In studies focusing exclusively on contact sports, the effect sizes tended to increase as study precision increased. In contrast, studies examining a mix of contact and non-contact sports reported more stable effect sizes that were closer to the null. This finding may reflect the inherent differences in risk associated with sustained exposure to high-impact collisions across various sporting environments.

No significant moderating effects were found for athlete level ($p = 0.130$), study design ($p = 0.148$), follow-up duration ($p = 0.629$), or country ($p = 0.590$). Despite the lack of statistical significance, visual inspection of the regression plots suggested a potential trend where effect sizes from retrospective cohort studies increased with greater precision, while those from prospective studies remained relatively constant. Similarly, effect estimates for different athlete levels, follow-up durations, and countries demonstrated minor fluctuations around the null effect line, with no clear or consistent patterns observed **(Figure 6).**

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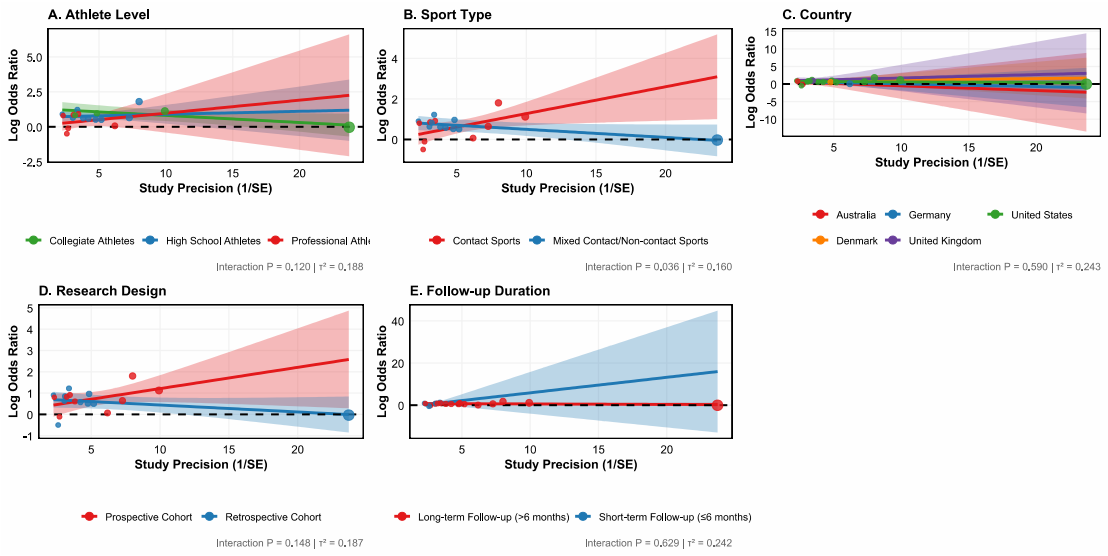

Figure 6: Meta-regression analysis of the risk of all-cause injury after concussion, stratified by study-level moderators.

3.10 Publication Bias

To evaluate potential publication bias, a funnel plot was constructed and Egger's regression test was performed. The funnel plot appeared roughly symmetrical **(Figure 7)**, indicating that smaller studies did not produce notably divergent effect sizes. This was supported by Egger's test, which showed no statistically significant asymmetry (p =0.0512); while this value approaches the threshold of significance, it suggests no strong evidence of publication bias. However, these findings should be interpreted with caution, acknowledging the limited statistical power of asymmetry tests to detect bias, particularly given the number of included studies.

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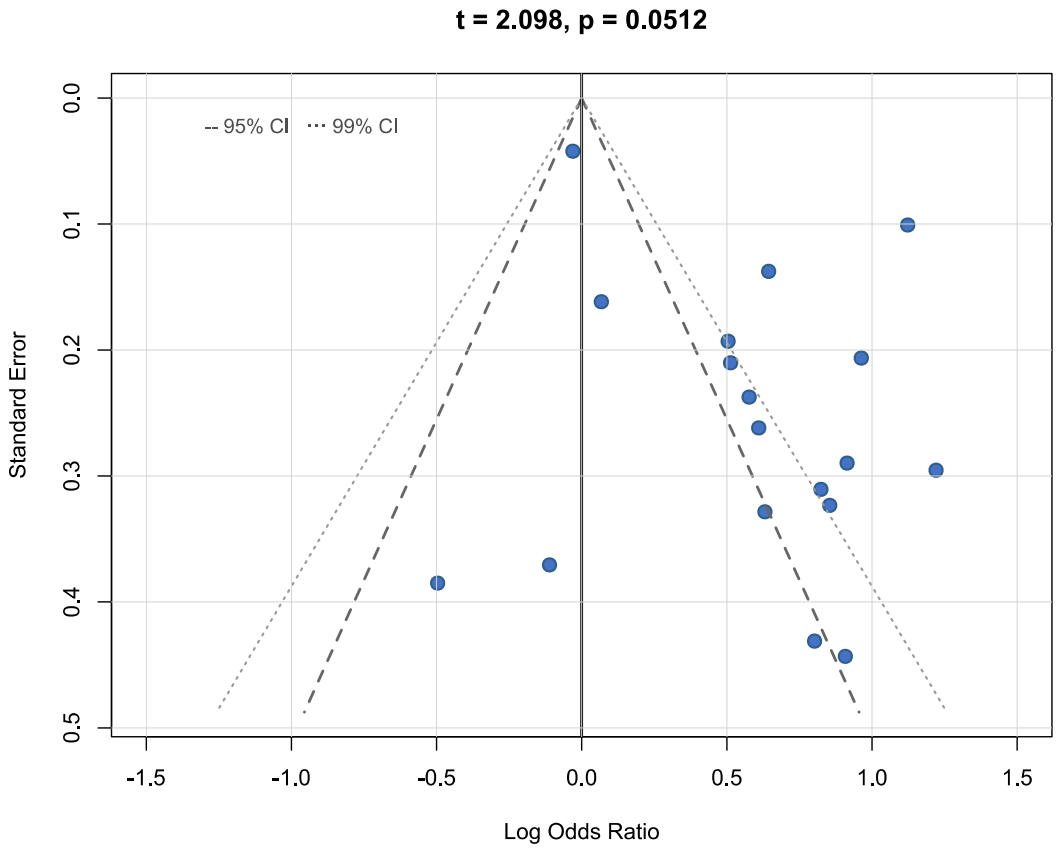

Figure 7 Publication bias of the risk of all-cause injury in athletes after concussion

4.DISCUSSION

4.1 Principal findings

This meta-analysis, encompassing 19 cohort studies and 86,879 athletes, provides robust evidence that a history of concussion is a significant risk factor for subsequent injuries. Our primary finding reveals that athletes who have sustained a concussion are nearly twice as likely to experience a subsequent all-cause injury compared to their non-concussed counterparts (OR = 1.93, 95% CI: 1.39－2.68). The risk was most pronounced for recurrent concussions, with previously concussed athletes facing a threefold greater odds of sustaining another concussion (OR = 3.06, 95% CI: 1.81－5.17).

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Furthermore, the risk extended to other body regions, with significantly elevated odds for both upper extremity (OR = 1.76, 95% CI: 1.10 – 2.81) and lower extremity injuries (OR = 1.49, 95% CI: 1.06 – 2.09).

4.2 Comparison with Prior Evidence and Novel Insights

The principal findings of this meta-analysis are consistent with and build upon the existing body of literature, strengthening the evidence base for the association between concussion and subsequent injury risk. Our pooled estimate for all-cause injury (OR=1.93) aligns remarkably well with the landmark meta-analysis by Mcpherson et al (2019)[36], which reported that concussed athletes had over two times greater odds of sustaining a subsequent musculoskeletal injury (MSKI) (OR=2.11). Similarly, our finding of a 49% increased risk for lower extremity injury (OR=1.49) is comparable to other recent systematic reviews. For instance, Jildeh et al (2022)[37] reported an odds ratio of 1.85 for LE injury within one year of concussion, and Mccann et al (2022b)[38] found a 58% greater risk (RR=1.58) specifically in college athletes. Furthermore, the threefold increased risk for recurrent concussion (OR=3.06) strongly corroborates the well-established principle that a prior concussion is the single greatest predictor for a future one [39]. This estimate is quantitatively similar to the findings of van Ierssel et al (2021)[40], who reported a pooled risk ratio of 3.64 for recurrent concussion in children and adolescents.

Beyond corroborating previous work, the present study offers several novel insights derived from our detailed subgroup and meta-regression analyses. A

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key new finding is the identification of sport type as a significant moderator of all-cause injury risk (meta-regression p=0.038). This suggests that the risk profile for athletes in contact sports may differ from that of athletes in cohorts that include participants from non-contact sports, potentially reflecting differences in exposure to high-impact forces or sport-specific movement patterns. Our subgroup analysis also revealed a nuanced picture across competition levels. The risk was most pronounced and statistically significant among college athletes (OR=2.29), whereas the estimates for professional (OR=1.40) and high school athletes (OR=2.03) did not reach statistical significance. This heterogeneity in findings across levels is not a contradiction but a critical insight. The strong signal in collegiate sports may be attributable to the high-quality, systematic injury surveillance systems common in large NCAA programs, which reduce measurement error and increase statistical power. Conversely, the non-significant finding in professional athletes could be multifactorial, stemming from superior, individualized rehabilitation, a "survivor bias" where only the most robust athletes compete [41], or smaller study sample sizes. At the high school level, the lack of statistical significance is likely due to methodological noise, as inconsistent injury reporting and high variability in athlete maturation and care introduce significant heterogeneity (I^2 =94.4%), obscuring a clear statistical signal.

The study also explored how injury risk evolves over time. While prior work suggested increased risk within 90 days [42] , our analysis indicates that the

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increase in risk was statistically significant in studies with follow-up periods exceeding six months (OR=1.94), whereas the association did not reach statistical significance in studies with shorter follow-up durations (<6 months; p=0.087). It is important to note that this is an observational finding. It suggests that within existing cohort studies, susceptibility to injury following concussion may not resolve immediately upon the cessation of short-term clinical symptoms, but rather remains observable over a more extended temporal window. This implies that the underlying neuromuscular deficits driving this elevated risk are not transient but persistent, manifesting as injuries during prolonged athletic exposure, thereby challenging the traditional notion of "rapid recovery." The prevailing hypothesis regarding the mechanism of this increased risk is that subtle, sub-clinical neurological deficits persist in athletes even after they are clinically deemed "recovered" following symptom resolution and standard return-to-play assessments [43]. These deficits include impaired neuromuscular and postural control, where concussion disrupts the brain's processing of visual, vestibular, and somatosensory information, leading to altered gait, diminished stability, and delayed reaction times [44]. Given that traditional clinical evaluations often fail to detect these subtle functional impairments, the Amsterdam International Consensus Statement and recent studies strongly advocate for the implementation of objective, standardized assessment tools [45]. For instance, the ImPACT test can be utilized to detect delays in cognitive reaction time, while the Sway or Balance Error Scoring

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System (BESS) serves to assess postural stability [46, 47]. Relying solely on symptom resolution without validation from these objective tools may fail to identify latent neuromuscular deficits that are sufficient to increase the risk of subsequent injury. Furthermore, altered sensorimotor integration and cognitive deficits may leave athletes with insufficient cognitive resources to maintain proper movement mechanics during complex competitive scenarios, thereby exacerbating injury risk [48].

4.3 Clinical and Preventive Implications

Recovery from concussion often extends beyond the resolution of symptoms, with ongoing neuromuscular and balance deficits that increase the risk of re-injury[49, 50]. These findings support extending post-concussion monitoring beyond six months and adopting individualized rehabilitation plans focused on restoring motor control, balance, and cognitive-motor integration[6, 51, 52]. High-risk groups—especially professional athletes—may benefit from interventions such as biomechanical screening and neurocognitive-motor training[53, 54]. Comprehensive assessment strategies that include physical, cognitive, and psychosocial evaluations can guide more effective treatment plans[55]. Recent evidence also suggests that early reintroduction of physical activity, rather than prolonged rest, may promote faster recovery[56, 57]. Proven approaches include aerobic exercise[58], cognitive rehabilitation, and vestibular therapy[59]. Return-to-play (RTP) decisions should account for the increased risk of MSK injury[60]. Incorporating neuromechanical assessments into RTP protocols can help ensure athletes are physically ready to resume competition. Ongoing injury

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surveillance—particularly in contact sports—along with educational programs promoting symptom reporting, is essential to preventing reinjury and addressing underreporting of concussion[61].

5. STRENGTHS, LIMITATIONS, AND FUTURE DIRECTIONS

The credibility of this meta-analysis is supported by several methodological strengths. The study was conducted in strict adherence to the PRISMA 2020 guidelines and was pre-registered with PROSPERO (CRD420251044641), ensuring methodological transparency and minimizing the risk of reporting bias. A comprehensive search strategy was executed across five major electronic databases, supplemented by manual reference checks, to ensure a thorough and representative sample of the available literature. The inclusion of 19 cohort studies with a large pooled sample of 86,879 athletes provides substantial statistical power and enhances the generalizability of the findings. Furthermore, we employed robust statistical methods, including a random-effects model appropriate for heterogeneous data, and conducted leave-one-out sensitivity analyses that confirmed the stability of our results. Crucially, by using subgroup and meta-regression analyses, this study moved beyond a simple pooled estimate to actively investigate potential sources of heterogeneity. Finally, formal testing with Egger's regression (p=0.051) and visual inspection of the funnel plot revealed no significant evidence of publication bias, further strengthening confidence in the results.

Despite the aforementioned strengths, several limitations must be

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acknowledged. First, a high degree of statistical heterogeneity was observed in the analysis of all-cause injury (I^2 = 94.4%). Although meta-regression identified sport type as a significant moderator, residual heterogeneity suggests that the pooled effect sizes should be interpreted as average associations across varying study contexts, rather than as uniform predictive values applicable to all settings. Second, while nearly three-quarters (73.7%) of the included studies adjusted for critical confounders such as prior injury history—and sensitivity analyses confirmed the stability of the results—the inclusion of unadjusted studies (n=5) and inconsistencies in covariate selection across studies may still introduce potential bias. Third, the robustness of evidence for certain subgroups is limited; the assessment of upper extremity injury was based on only two studies, and the lack of significant risk observed in professional and high school athlete subgroups is more likely attributable to insufficient statistical power or methodological discrepancies rather than an absence of risk; thus, it should not be inferred that these populations are risk-free. Finally, heterogeneity in the definitions of "concussion" and "injury," potential unmeasured confounders, and language bias resulting from restricting the search to English-language literature may affect the precision of the effect estimates. Future research should prioritize large-scale, multinational, prospective cohort studies utilizing standardized definitions to further elucidate causal relationships and optimize prevention strategies.

6.CONCLUSION

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In conclusion, this systematic review and meta-analysis provides robust evidence that athletes with a history of concussion face a significantly elevated risk of sustaining various subsequent injuries. The findings explicitly indicate that symptom resolution does not equate to full functional recovery. Although athletes may appear symptom-free, objective assessment tools—such as the ImPACT or Sway tests—may reveal persistent neuromuscular deficits that do not fully resolve despite the cessation of reported symptoms. Consequently, relying solely on symptom clearance as the benchmark for "clinical recovery" may fail to accurately reflect an athlete's true functional status. A window of heightened vulnerability to injury persists for a prolonged period after athletes are cleared to return to play under current standard protocols. Accordingly, these results support the refinement of existing concussion management protocols, particularly the integration of scientifically validated, objective functional assessments, to ensure full recovery prior to return-to-play and to mitigate the risk of subsequent injury. Adopting such a comprehensive strategy is essential for safeguarding the short- and long-term health of athletes.

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Abbreviations

MeSH Medical Subject Headings

PRISMA Preferred reporting items for systematic reviews and meta-analyses

RTP Return-to-play

Supporting information

Tables 1~5. The literature search steps and results of PubMed, Embase, the Cochrane Library, Web of Science, and EBSCO.

Fig 1 Sensitivity analysis of the risk of all-cause injury in athletes after concussion

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Fig 2 Sensitivity analysis of the risk of upper extremity injury in athletes after concussion

Fig 3 Sensitivity analysis of the risk of lower extremity injury in athletes after concussion

Fig 4 Sensitivity analysis of the risk of concussion injury in athletes after concussion

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Author contributions

MYL conceived and designed the research project. MYL and YG searched and screened the studies. All authors carried out the risk of bias assessment and data extraction. KYL analysed the data and interpreted the results of the analysis. MYL produced the initial draft of the manuscript. All authors drafted the manuscript and assisted in the interpretation of the data. All authors critically reviewed the manuscript, provided significant input to the initial submission and subsequent revisions, and approved the final version. KYL is responsible for the overall content as the guarantor.

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Availability of Data and Materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics Approval and Consent to Participate

Not applicable.

Consent for Publication

Not applicable.

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Competing interests

MYL, YG, and KYL declare that they have no competing interests.

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